Aayush Rai

BS-MS in Mathematics and Computing, IIT Roorkee
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EDUCATION

Indian Institute of Technology, Roorkee Aug 2023 - May 2028
BS-MS in Mathematics and Computing *CGPA : 8.825*

St. George's Preparatory School Apr 2022 - Mar 2023
Intermediate (Class XII) *Percentage : 92.40%*

Nagaji Saraswati Vidya Mandir Apr 2020 - Mar 2021
Matriculate (Class X) *Percentage : 91.60%*

PROJECTS

Multithreaded File Server with Access Control | Github Jan 2026 - Apr 2026
Self Project
- Built a high-performance **file server** in **C++** supporting concurrent client requests through **multithreading**, synchronization primitives, and **TCP/IP networking**.
- Implemented a **thread-safe architecture** using POSIX threads and reader-writer locks, handling **100+** simultaneous connections while ensuring efficient file access, process synchronization, authentication, and role-based access control.

Design and Implementation of a Memory Management Simulator | Github Dec 2025 - Jan 2026
- Built a **C++ simulator** modeling modern OS memory management, including **dynamic allocation**, **virtual memory**, and **multi-level cache hierarchies** for performance analysis and educational system modeling.
- Implemented **First-Fit**, **Best-Fit**, **Worst-Fit**, and **Buddy allocators** with block splitting/coalescing, paging-based virtual memory, L1/L2 cache simulation (FIFO/LRU), and real-time fragmentation analytics.

DocuNova: Smart PDF Reader with AI-Powered Outline & Insights | Website Github Jul 2025 - Sep 2025
Vision and Language Group, IIT Roorkee
- Built a **Retrieval-Augmented Generation (RAG)** system for large-scale PDF intelligence, combining **semantic retrieval**, **vector embeddings**, and **LLM-powered document understanding**.
- Developed persona-aware retrieval and ranking pipelines with document chunking, embedding-based search, multi-document reasoning, citation-grounded generation, and structured PDF extraction using Adobe Embed API.

E-Waste Screening System | Github Jan 2024 - Mar 2024
MaRS (Models and Robotics Society), IIT Roorkee
- Real-time embedded vision system for automated e-waste detection and classification across categories like mobile phones, PCBs, and batteries targeting intelligent waste management and recycling automation pipelines.
- Applied **Transfer Learning** with **ResNet** & **VGG** achieving **93.4%** accuracy; fine-tuned **YOLOv9** with FPN for real-time bounding box prediction achieving **97.9%** mAP across e-waste categories.

AWARDS & ACHIEVEMENTS

- Secured **All India Rank 2758** in **JEE Advanced, 2023** among **250,000** top qualifiers across India.
- Secured **All India Rank 4140** (**99.65 percentile**) in **JEE Mains, 2023** out of over **1.2 million** candidates.
- Achieved a **Top 100 National Ranking** among over **1 Lakh Teams** in the **Adobe India Hackathon** competing as **Team IItr**.
- Secured **Global Rank 782** in Meta Hacker Cup 2025 Round 2 and Global Rank 2183 in Round 1.
- Secured **Global Rank 635** in Codeforces Round 1080.

SKILLS

Programming Languages : C++, Python
Additional Courses : Data Structures and Algorithms, Fundamentals of Object Oriented Programming, Data Science

POSITIONS OF RESPONSIBILITY & EXTRA CURRICULARS

Competitive Programmer Online Judges
Oct 2024 - Present
- Attained **Codeforces Expert** (Max Rating: **1724**, handle: **AntiMaTTeR_007**) and **CodeChef 4-Star Coder** (Max Rating: **1818**, handle: **aayu_iitr**), demonstrating strong algorithmic problem-solving and data structure expertise.

Coordinator Placement and Internship Cell (PIC), IIT Roorkee
Aug 2025 - Feb 2026
- Coordinated placement and internship recruitment operations for **20+** companies during the placement season, managing company interactions, candidate coordination, and seamless drive execution.

Executive Member Team Wellness, IIT Roorkee
Sep 2025 - Present
- Coordinated and supported student well-being initiatives, including the successful organization of the **Inter IIT Wellness Meet 2026** at IIT Roorkee, fostering engagement, inclusivity, and mental health awareness.

REFERENCES

Dr. Sanjeev Kumar, Head of the **Department of Data Science and Artificial Intelligence**, Indian Institute of Technology, Roorkee
Dr. Maheshanand, Head of the **Department of Mathematics**, Indian Institute of Technology, Roorkee